Chemistry 3A – Spring 2026 RETAKE VERSION

Sections 19320-19321

Midterm Examination #1

Indicate **BY DIRECTLY CIRCLING** CLEARLY on the exam page *one* and *only one* choice that is **BEST** response for the question. Point values: multiple choice 3 pt; true/false 1 pt

1. Which element belongs to the **alkali metals** group?
   (a) calcium **(b) sodium** (c) magnesium (d) chlorine

LS2 slide 25

2. Which subatomic particle has a mass of approximately **1 amu** and a positive charge?
   (a) electron **(b) proton** (c) neutron (d) none of the above

LS2 slides 11, 14

3. Which of the following is a **polyatomic ion**?
   (a) Cl^- (b) Na^+ **(c) SO_4^{2-}** (d) O^{2-}

4. Convert 25.0 °C into Kelvin.
   (a) 0 K (b) 25 K (c) 273 K **(d) 298 K**

LS1, slides 47-51

5. A sample has a mass of 12.5 g and a volume of 2.50 mL. What is its density?
   (a) 0.200 g/mL **(b) 5.00 g/mL** (c) 31.3 g/mL (d) 0.050 g/mL

LS1, slides 41-46

6. Which of the following is the correct name for $Ca(NO_3)_2$?
   (a) cobalt fluoride (b) calcium nitroxide **(c) calcium nitrate** (d) calcium oxide

7. Which of the following terms describes the distance between two wave crests?
   (a) wave speed (b) frequency **(c) wavelength** (d) amplitude

LS3, slide 4

8. Which of the following is the correct atomic notation for carbon-14?
   (a) ${}^{12}_{6}C$ **(b)** ${}^{14}_{6}C$ (c) ${}_{14}^{6}C$ (d) ${}^{60}_{27}Co$

LS8 slides 11-14

9. Convert 4.56×10^{-3} into standard notation
   (a) 0.00456 (b) 4560 (c) 0.000456 (d) 4,560

LS1, slide 11: verification of a number in scientific notation

10. Which of the following is the electron configuration for hydrogen?
    (a) **$1s^1$** (b) $1s^2$ (c) $1s^2 2s^1$ (d) $1s^2 2p^1$

LS3, slides 28-29

11. Which of the following is TRUE about element bromine (Br)?
(a) It is a member of the *d*-block of elements
(b) It is a Group 1 element
(c) It is a non-metal
(d) none of the above

LS2, slide 27. This element shown in the **halogens** group. This rules out choice (b) since Group 1 elements are the alkali metals, and bromine is not a metal. The d-block elements do not include the Group 17 elements, of which bromine is a member. Choice (c) is true, so choice (d) cannot be true.

12. What term describes an electron normally in n = 1 shell acquiring energy to be in n=2 shell?
(a) emission (b) excited state (d) ground state **(d) absorption**

LS3, slide 15

13. Express 0.0000721 in scientific notation with correct significant figures.
(a) 7.21×10^{-4} **(b) 7.21×10^{-5}** (c) 7.21×10^{-6} (d) 7.21×10^{-7}

LS1 slides 6-16

14. When you complete the operation on the expression 1.450 × 3.01 / 10.025, how many significant digits will the final result have? ('/' = division operation)
(a) 2 **(b) 3** (c) 4 (d) 5

LS1, slide 24

15. The scientific method includes a stage where a **tentative explanation** is given to account for a set of related observations. What is that stage?
(a) law **(b) hypothesis** (c) theory (d) experimentation

LS2, slide 5

16. Which of these is a polyatomic ion?
(a) Ne **(b) CO_3^{2-}** (c) Br^- (d) Na^+

17. Which of these is the ground-state electron configuration for the **carbon (C)** atom?
(a) $1s^2 2s^2 2p^6$ **(b) $1s^2 2s^2 2p^2$** (c) $1s^2 2s^2 2p^4$ (d) $1s^2 2s^2$

LS3, slide 29. It is not necessary to memorize electron configurations but to understand where the orbital types are in the Table, and to make sure that all the superscript numbers add up as a count of electrons that is equal to the element's atomic number. Carbon has atomic number 6, so it should have 6 total electrons, which is what choice (b) gives. It is also only necessary to remember the range of superscripted numbers for the orbital type: *s*: s^1-s^2, *p*: p^1-p^6, *d*: d^1-d^{10}, *f*: f^1-f^{14}.

18. How many quantum numbers are required to uniquely define the state of an electron in an atom?
(a) 1 (b) 2 (c) 3 **(d) 4**

LS3 slide 19

19. Perform the operation: 3.456 + 12.1 + 0.0045. Report with correct decimal places.
(a) 15.5605 (b) 15.56 **(c) 15.6** (d) 15.561

LS1, slide 23. Start by entering everything in the calculator and you get on your calculator display: "15.5605". This is an addition operation entirely, do the decimal place rules apply. Choice (a) is just the calculator display for those who do not know the decimal place rules. Choices (b) and (d) are there for those still not understanding the rules.

20. The label on the bottle has the formula $FeCl_3$. What is its name using the Stock system?
(a) iron chloride (b) iron(II) chloride **(c) iron(III) chloride** (d) iron monochloride

21. One of the quantum numbers of an electron is related to the shape of the electron's orbital and is represented by a letter in this set: {*s*, *p*, *d*, *f*}. Which quantum number is this?
(a) *n* **(b) l** (c) m_s (d) Z

LS3 slide 19

22. Which of these subatomic particles is **NOT** found in the nucleus of the atom?
(a) proton (b) neutron **(c) electron** (d) none of the choices (a) to (c)

LS2, slide 11

23. What is a TRUE statement regarding an element in the Periodic Table?
(a) the number of neutrons and protons for every element must be equal
(b) the atomic number (Z value) indicates the number of protons it has
(c) the mass number (A value) indicates the number of electrons and protons it has
(d) the isotope of the element is determined by the number of electrons

LS2 multiple slides

Nothing in lecture content stated that the number of protons and neutrons must be equal in the nucleus for the elements. In fact, atomic nuclei show instability (a radioactive nature) when protons and neutrons are equal. Choice (a) is not true.

Choice (c): the mass number (A) was well defined: it is the number of protons and neutrons in the nucleus, and "electrons" were not part of the definition, so this choice is false

Choice (d): The isotope of an element is basically determined by its mass number (A value), so this is false.

Choice (b) is the definition of Z, so this choice is true

24. What are the Group 2 elements called?
(a) Noble gases (b) Alkali metals **(c) Alkaline Earth metals** (d) Halogens

LS2, slide 26

25. An isotope of an element has atomic number (Z) = 15 and mass number (A) of 31. How many neutrons are in the nucleus?
(a) 15 **(b) 16** (c) 31 (d) 46

LS2 slide 15. Z = protons, A = protons + neutrons. So neutrons = $A - Z = 31 - 15 = 16$

Circle either TRUE or FALSE whether the statement is true or false.

26. A **cation** is an ion that has a net negative charge because it gained one or more electrons.
True **False**

27. Ionic compounds with monatomic elements are typically formed between metals and nonmetals.
True False

28. The speed/velocity of a electromagnetic wave can be calculated by dividing the frequency by the wavelength: $c = \nu / \lambda$
True **False**

LS3 Slide 5: the equation shows the wavelength MULTIPLIED by frequency equals the speed of the wave, so doing the algebra shows that is wrong.

29. The **p-block** of elements in the Periodic Table starts in Period 1.
True **False**

LS3, slide 31. The p orbitals are filled with their first electron with the element boron, which is in Period 2. Period 1 has only two elements, hydrogen and helium, and they only have s orbitals

30. The **Pauli Exclusion Principle** states that an orbital can have only two electrons which are paired up with opposite spins
True False

LS3, slide 26

31. According to the rules for significant figures, **conversion factors** (like 1 m = 100 cm) are considered to have infinite significant digits and do not affect the precision of a calculation.
True False

LS1, slides 32-40

32. The atomic number (*Z*) represents the number of protons in an atom.
True False

LS2, slide 15

33. The only subatomic particles allowed in the nucleus of the atom are **protons**.
True **False**

LS2, slides 11, 13, 14

34. Noble gases have complete valence shells, which is why they do not form compounds with other elements.
True False

LS2, slide 28. Noble gases of the Group 18 elements are called "noble" because no other element "touches them" by forming compounds with these elements. They are considered "inert" and "unreactive". (There are exceptions with higher period noble gases now found, but for the most part, these are unreactive.)

35. The law of conservation of mass states that matter **CAN** be created or destroyed in chemical reactions.
True **False**

The principle of the conservation of mass has been mentioned many instances in the semester, with the principle that mass (as matter) cannot be created or destroyed but that it may change form particularly through chemical reactions. The principles of conservation of energy and of electric charge are also principles.